

SEC P vs NP log 2026-05-21

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-21 09:45 UTC

SEC P vs NP — daily log 2026-05-21

26 cycles recorded

Noncommutative Algebraic Growth and Tseitin Resolution Width

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative Algebra $\times$ Resolution Width of Tseitin Formulas
- **Recorded:** 2026-05-21 00:30 UTC
- **Entry ID:** e87da145bc32

Statement

For a random 3-CNF formula Φ with n variables, let A_Φ be the noncommutative algebra generated by variables x_i and relations $x_i x_j = x_j x_i$ if clauses (i,j) are satisfied. The growth rate of $\dim(A_\Phi_k)$ is $\Omega(2^{\lfloor k/2 \rfloor})$ for all $k \leq \log n$ iff Φ 's Tseitin formula has resolution width $\geq \log n$.

Rationale

Noncommutative algebras encode clause interactions via generators, while resolution width measures proof complexity. The conjecture links algebraic growth rates to proof complexity via commutativity constraints, revealing structural trade-offs between algebraic dependencies and proof efficiency.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 27.65s

```
0.0, 'instances_tested': 10, 'conjecture_holds': False, 'counterexample': 'growth_rate_mismatch'}
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds': False}
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```

```
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds': True}
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds': True}
RESULT: FALSIFIED counterexample="growth_rate_mismatch" first_failing_seed=0
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: The conjecture was falsified by a counterexample showing growth_rate_mismatch in all tested instances. | next: Analyze the specific counterexample to determine how the growth rate deviation correlates with Tseitin resolution width properties

Noncommutative Rank Gap in Read-Twice Branching Programs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative Algebra $\times$ Read-Twice Branching Programs
- **Recorded:** 2026-05-21 00:52 UTC
- **Entry ID:** f245272528b8

Statement

For a read-twice BP P over n variables, the noncommutative rank of its transition matrix M satisfies $\text{rank}_{\text{nc}}(M) \geq \Omega(n)$, whereas for read-once BPs, $\text{rank}_{\text{nc}}(M) = O(\log n)$.

Rationale

Noncommutative rank captures algebraic dependencies in matrix factorizations, which may reflect the repeated variable access in read-twice BPs. This could expose structural asymmetries in their computational power compared to read-once BPs.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 105, in <module>

trial_result = run_trial(seed)
^^^^^^^^^^^^^^

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 78, in run_trial

rank = rank_nc(M, p)
^^^^^^^^^^^^^^

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 38, in rank_nc

gaussian_elimination(A, p)

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 23, in gaussian_elimination

if abs(A[j][i]) > abs(A[max_row][i]):
^^^^^^^^^^^^^^

TypeError: bad operand type for abs(): 'list'

Judge reasoning

Test crashed due to TypeError in Gaussian elimination, preventing result collection. Support condition cannot be evaluated without successful trials. | next: Debug Gaussian elimination implementation to handle matrix entries correctly (ensure numerical values instead of lists)

Non-Abelian Harmonic Analysis and SOS Refutation Degree for CSPs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Non-Abelian Harmonic Analysis × SOS Refutation Degree for CSPs
- **Recorded:** 2026-05-21 01:05 UTC
- **Entry ID:** 93e2fe31dd48

Statement

For a CSP instance over a non-abelian group G , the SOS refutation degree is bounded by the dual norm of the Fourier coefficients of the CSP's constraint function. Specifically, for every CSP instance φ over G , $\text{refutation_degree}(\varphi) \leq \|F\|_\infty$, where F is the Fourier transform of φ 's constraint function.

Rationale

Non-abelian harmonic analysis decomposes CSP constraints into group characters, revealing structural symmetries that may influence the SOS hierarchy's ability to certify unsatisfiability. The dual norm of Fourier coefficients quantifies the 'spread' of constraints across irreducible representations, which could correlate with the minimal SOS degree required for refutation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c3843cd.py", line 74, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c3843cd.py", line 50, in run_trial
    F = [1] * (2 ** n) # Placeholder Fourier coefficients
         ^^^^^^^^^^^
```

MemoryError

Judge reasoning

Test crashed with MemoryError before producing results, preventing evaluation of support fraction or counterexamples. | next: Optimize memory usage in test code for large n , then re-run experiments

Non-Abelian Fourier Coefficient Gap Bounds Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE

- **Bridge:** Non-Abelian Harmonic Analysis × Communication Complexity of Disjointness
- **Recorded:** 2026-05-21 01:53 UTC
- **Entry ID:** 324ca13de3e5

Statement

For any 3-CNF formula ϕ with n variables, the minimal non-Abelian Fourier coefficient $|F[\pi]|$ over all irreducible representations π of the symmetric group S_n satisfies $|F[\pi]| \geq \Omega(1/\sqrt{n})$ if and only if the disjointness communication complexity of ϕ is $\Omega(\log n)$.

Rationale

Non-Abelian Fourier analysis captures group-theoretic symmetries in communication matrices. Disjointness requires coordination across independent sets, which may manifest as nontrivial Fourier coefficients in symmetric group representations, revealing structural constraints on protocol efficiency.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6de39bfc.py", line 65, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6de39bfc.py", line 45, in run_trial
    clause = [random.randint(0, n-1) for _ in range(3)]
              ^
```

NameError: name 'n' is not defined

Judge reasoning

Test crashed due to undefined variable 'n' before producing results | next: Fix the NameError in test code by defining 'n' before using it in clause generation

ϵ -Approximation Complexity Lower Bound for k-CLIQUE via Monotone DNF Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Approximation Theory of Boolean Functions × Monotone DNF Formulas for k-CLIQUE
- **Recorded:** 2026-05-21 02:10 UTC
- **Entry ID:** 5c00c6af4f62

Statement

For any $\epsilon \in (0, 1/2)$, the ϵ -approximation complexity of the k-CLIQUE function is $\Omega(n)$ when $k \geq \log n$, and this complexity is submodular under conjunction of functions.

Rationale

Approximation theory provides tools to analyze how well boolean functions can be approximated by simpler functions, which could reveal structural limitations in monotone DNF representations. The submodularity under conjunction suggests that combining functions increases the required complexity, aligning with known lower bounds for k-CLIQUE.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 70, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 49, in run_trial
    clique = generate_k_clique(n, k)
             ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 25, in generate_k_clique
    if all(node not in edge for edge in clique):
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 25, in <genexpr>
    if all(node not in edge for edge in clique):
        ^^^^^^^^^^^^^^^^^
```

TypeError: argument of type 'int' is not iterable

Judge reasoning

Test crashed with TypeError due to non-iterable integer in membership check, preventing data collection | next: Fix the generate_k_clique function to ensure edges are iterable structures (e.g., tuples) before membership testing

Matroid Rank and ABP Size for Polynomial Representations

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid Theory × Algebraic Branching Program Size
- **Recorded:** 2026-05-21 02:20 UTC
- **Entry ID:** d2edbef09287

Statement

The matroid rank of the coefficient matrix of a polynomial represented by an ABP of size s is $\Theta(\log s)$.

Rationale

Matroid rank captures the linear independence structure of the polynomial's coefficients, which may reflect the minimal complexity of representing the polynomial via an ABP. This bridge could expose hidden algebraic dependencies in the ABP's structure.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2b11b003.py", line 75, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2b11b003.py", line 51, in run_trial
    A = [[coefficients[i] * j**k for k in range(n + 1)] for i in range(n + 1)]
            ^
```

NameError: name 'j' is not defined

Judge reasoning

Test crashed due to undefined variable 'j', preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the NameError in test_2b11b003.py by defining variable 'j' in the list comprehension

Sheaf Cohomology Euler Characteristic and Tseitin Resolution Proof Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Sheaf Cohomology $\times$ Resolution Proof Size for Tseitin Formulas
- **Recorded:** 2026-05-21 02:31 UTC
- **Entry ID:** 62b0c1d5061d

Statement

For a Tseitin formula derived from a graph G with n vertices, the absolute value of the Euler characteristic of the sheaf cohomology groups $H^i(G)$ is inversely proportional to the resolution proof size, i.e., $|\chi(H^i(G))| = \Theta(1 / \log n)$.

Rationale

Sheaf cohomology captures the topological structure of the graph, which may influence the complexity of the resolution proof. The Euler characteristic provides a quantitative measure of this topological invariant, potentially linking it to the proof size.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ff24d2c.py", line 113, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ff24d2c.py", line 43, in run_trial
    clauses.append([-literals[i], -literals[j]])
                    ^^^^^^^^^^^^^
```

TypeError: bad operand type for unary -: 'str'

Judge reasoning

Test crashed due to type error in clause generation, preventing data collection | next: Fix the Type-Error by ensuring literals are integers before applying unary negation

Fourier Norm Lower Bound for Disjointness Communication Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Harmonic Analysis on Locally Compact Groups × Randomized Communication Complexity of Disjointness
- **Recorded:** 2026-05-21 03:12 UTC
- **Entry ID:** af4613884bf0

Statement

For any $n \times n$ communication matrix M of the Disjointness problem, the L^2 norm of its Fourier transform over the additive group $\mathbb{Z}_n^2$ is $\Omega(n)$.

Rationale

Harmonic analysis on finite abelian groups captures the spectral structure of communication matrices, with Fourier norms reflecting the ‘spread’ of information across frequencies. The Disjointness matrix’s inherent combinatorial structure may enforce nontrivial Fourier norms, linking algebraic invariants to communication complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 70, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 54, in run_trial
    F = fft_2d(M)
        ^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 43, in fft_2d
    M_fft[i] = fft([M[i][j] for j in range(n)])
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 34, in fft
    even = fft([a[k] for k in range(0, n, 2)])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 34, in fft
    even = fft([a[k] for k in range(0, n, 2)])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 34, in fft
    even = fft([a[k] for k in range(0, n, 2)])
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
[Previous line repeated 2 more times]
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 36, in fft
    T = [cmath.exp(-2j * math.pi * k / n) * odd[k] for k in range(n // 2)]
    ^^^^^
NameError: name 'cmath' is not defined. Did you mean: 'math'? Or did you forget to import 'cmath'?
```

Judge reasoning

Test crashed due to missing cmath import, preventing evaluation of Fourier norm bounds. | next:
Fix the test by importing cmath and re-running with proper error handling

Laplacian Eigenvalue and Resolution Length of Tseitin Formulas

- **Verdict:** INCONCLUSIVE
- **Bridge:** Spectral Graph Theory × Resolution Length of Tseitin Formulas
- **Recorded:** 2026-05-21 03:30 UTC
- **Entry ID:** 9e9baa385874

Statement

For any graph G , the resolution length of the Tseitin formula on G is at least $2^{\Omega(1/(1 - \lambda_2))}$, where λ_2 is the second smallest eigenvalue of the Laplacian matrix of G . For graphs where $1 - \lambda_2 \geq c$ (constant $c > 0$), the resolution length is exponential in $1/(1 - \lambda_2)$.

Rationale

The second eigenvalue of the Laplacian (λ_2) captures the expansion properties of the graph. A smaller λ_2 indicates a better expander, leading to longer resolution proofs as the expansion increases, requiring more steps to refute the Tseitin formula.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 85, in <module>
    result = run_trial(seed)
    ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 57, in run_trial
    λ2 = min(laplacian_eigenvalues(G)[1:])
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 41, in laplacian_eig
    L = gaussian_elimination(L)
    ^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 28, in gaussian_elim
    A[k][j] += factor * A[i][j]
    ~~~~^
```


IndexError: list index out of range

Judge reasoning

Test crashed with IndexError in Gaussian elimination, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the Gaussian elimination implementation to handle matrix indices properly and re-run tests with multiple seeds

Moment Matrix Spectral Gap and SOS Degree for Max-CUT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Random Matrix Theory $\times$ SOS Degree for Max-CUT
- **Recorded:** 2026-05-21 03:38 UTC
- **Entry ID:** 797c606cf901

Statement

For any n -variable Max-CUT instance, the smallest eigenvalue of its degree- d moment matrix satisfies $\lambda_{\min} \geq \Theta(1/\sqrt{n})$ iff $d \geq \Omega(\log n)$. The spectral gap $\gamma = \lambda_{\max} - \lambda_{\min}$ scales as $\gamma = \Theta(1/n)$ for $d = \Omega(\log n)$.

Rationale

Random matrix theory provides tools to analyze the spectral properties of moment matrices, which are central to SOS hierarchies. The spectral gap directly influences the convergence rate of SOS approximations, linking algebraic structure to algorithmic complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cld82cae.py", line 180, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cld82cae.py", line 153, in run_trial
    moment_matrix = construct_moment_matrix(instance, d)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cld82cae.py", line 39, in construct_mom
    m[i][j] += sum(lit ** abs(j - k) for lit in clause)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cld82cae.py", line 39, in <genexpr>
    m[i][j] += sum(lit ** abs(j - k) for lit in clause)
                   ^
```

NameError: name 'k' is not defined

Judge reasoning

Test crashed due to undefined variable 'k' in the moment matrix construction, preventing data collection. | next: Debug the NameError in the genexpr by checking variable scoping for 'k' in the clause summation

Free Entropy Gap in Read-Twice BP Transition Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability $\times$ Read-Twice Branching Programs
- **Recorded:** 2026-05-21 03:52 UTC
- **Entry ID:** b6ef2369c2bb

Statement

For a read-twice BP P over n -bit inputs, let μ_P be the non-commutative distribution of its transition matrices. Define $\rho(P) = |R(\mu_P)(0)|$, where $R(\mu_P)$ is the R -transform of μ_P . Then $\rho(P) \geq \Omega(\log n)$ for all P computing IP_2 , but $\rho(P) = O(1)$ for all read-once BPs computing IP_2 .

Rationale

Free entropy measures the 'non-commutative complexity' of random variables, which may capture the inherent structure of read-twice BPs that read-once BPs lack. The R -transform's zeroth coefficient encodes global scaling behavior, potentially distinguishing these models via their algebraic invariants.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 8.28s

```
e': Fraction(1, 1), 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'ρ(P) < 3.5'
TRIAL: {'metric_name': 'ρ(P)', 'metric_value': Fraction(1, 1), 'instances_tested': 30, 'conjecture_holds': False}
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TRIAL: {'metric_name': 'ρ(P)', 'metric_value': Fraction(1, 1), 'instances_tested': 30, 'conjecture_holds': False}
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9497589.py", line 76, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9497589.py", line 76, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~^
```

KeyError: 'seed'

Judge reasoning

Test crashed before producing reliable data, but stdout suggests counterexample $\rho(P) < 3.5$. The crash invalidates the counterexample's validity. | next: Re-run test with increased instance count and seed diversity to verify conjecture status

Matroid Rank Gap in Monotone DNF for k-CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid Theory × Monotone DNF Formulas for k-CLIQUE
- **Recorded:** 2026-05-21 04:29 UTC
- **Entry ID:** 202b457c781c

Statement

For a k-CLIQUE indicator function expressed as a monotone DNF with m clauses, the rank of the matroid formed by its clause vectors is $\Omega(n)$ if $m \leq n^{\{O(1)\}}$, and $O(\log n)$ if $m \geq n^{\{O(k\{1/4\})\}}$.

Rationale

Matroid rank captures combinatorial dependencies in clause interactions, which may expose structural barriers to succinct DNF representations. The conjecture ties this invariant to known lower bounds for k-CLIQUE, suggesting rank gaps could amplify hardness.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_115ac035.py", line 83, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_115ac035.py", line 56, in run_trial
    matrix = [[int(abs(lit) == var) for var in range(1, n + 1)] for clause in dnf]
               ^^^
```

NameError: name 'lit' is not defined. Did you mean: 'List'?

Judge reasoning

Test crashed due to NameError, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the NameError in test_115ac035.py by defining 'lit' variable before usage

Algebraic Shifting Facet Count Bounds Tseitin Resolution Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic shifting × Resolution length of Tseitin formulas
- **Recorded:** 2026-05-21 04:56 UTC
- **Entry ID:** 312ea3700c09

Statement

For a Tseitin formula on a graph G , the resolution length is $\geq 2^{\Omega(v(G))}$, where $v(G)$ is the number of facets in the algebraic shifted complex of G . This holds for all graphs with expansion parameter $\lambda(G) \geq 1/\sqrt{n}$.

Rationale

Algebraic shifting captures combinatorial structure of graphs via simplicial complexes, while resolution length measures proof complexity. The conjecture links structural properties of shifted complexes to proof complexity, leveraging the interplay between algebraic topology and proof complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91e2d45b.py", line 105, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91e2d45b.py", line 80, in run_trial
    expansion_param = sum(len(list(graph.neighbors(v))) for v in range(n)) / (n * (n - 1))
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91e2d45b.py", line 80, in <genexpr>
    expansion_param = sum(len(list(graph.neighbors(v))) for v in range(n)) / (n * (n - 1))
                        ^^^^^^^^^^^^^^^^^
```

AttributeError: 'set' object has no attribute 'neighbors'

Judge reasoning

Test crashed due to AttributeError in graph.neighbors() call on set objects, preventing data collection | next: Fix graph representation to use a structure with neighbors() method (e.g., NetworkX Graph) and re-run experiments

Symmetric Group Orbit Count Distinguishes Read-Twice BPs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Invariant Theory $\times$ Read-Twice Branching Programs
- **Recorded:** 2026-05-21 05:31 UTC
- **Entry ID:** 376a7658c1fe

Statement

For a read-twice BP P , the number of orbits of its associated polynomial representation under the symmetric group S_n action is $\Theta(\log \text{size}(P))$ for read-once BPs but $\Omega(n)$ for read-twice BPs. The orbit count is strictly greater for read-twice BPs when $\text{size}(P) \geq 2^{n/2}$.

Rationale

Invariant theory provides a framework to measure symmetry in polynomial representations, which could expose structural differences between read-once and read-twice BPs. The symmetric group action on variables creates a natural invariant that may scale differently with BP complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 245.34s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results; insufficient data to evaluate support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation

Polynomial Threshold Function Degree Bounds Karchmer-Wigderson Protocol Cost for PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Polynomial Threshold Functions $\times$ Karchmer-Wigderson Protocol Cost
- **Recorded:** 2026-05-21 05:41 UTC
- **Entry ID:** 8a180d66cac2

Statement

For any CNF formula F on n variables computing PARITY, the minimal degree of a polynomial threshold function representing F is $\Omega(\log n)$.

Rationale

Polynomial threshold functions capture the structure of Boolean functions, and their degree may reflect the complexity of communication protocols needed to compute them. The Karchmer-Wigderson protocol cost measures the communication complexity of computing a function, linking algebraic representation to interactive proof strategies.

Novelty

- Judge: ADJACENT_OK over 7 arXiv hits

Top hits consulted: - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1504.01064v2] The degree of the Alexander polynomial is an upper bound for the topological slice genus - [1806.07358v1] No Threshold graphs are cospectral - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer

Empirical Test

- exit code: 0, elapsed: 0.06s

```

: 1, "conjecture_holds": True, "counterexample": ""}
TRIAL: {"seed": 503, "metric_name": "polynomial_threshold_function_degree", "metric_value": 7.700825}
TRIAL: {"seed": 547, "metric_name": "polynomial_threshold_function_degree", "metric_value": 3.529875}
TRIAL: {"seed": 593, "metric_name": "polynomial_threshold_function_degree", "metric_value": 2.341542}
TRIAL: {"seed": 631, "metric_name": "polynomial_threshold_function_degree", "metric_value": 2.728451}
TRIAL: {"seed": 677, "metric_name": "polynomial_threshold_function_degree", "metric_value": 6.381222}
TRIAL: {"seed": 727, "metric_name": "polynomial_threshold_function_degree", "metric_value": 7.854839}
TRIAL: {"seed": 773, "metric_name": "polynomial_threshold_function_degree", "metric_value": 3.234422}
TRIAL: {"seed": 821, "metric_name": "polynomial_threshold_function_degree", "metric_value": 8.674023}
TRIAL: {"seed": 877, "metric_name": "polynomial_threshold_function_degree", "metric_value": 1.723483}
TRIAL: {"seed": 929, "metric_name": "polynomial_threshold_function_degree", "metric_value": 6.568578}
RESULT: SUPPORTED mean=4.618178297706176 std=2.272930228914458 support_fraction=0.9

```

Judge reasoning

parse_fail | next: NONE

Fourier Coefficient Sum Bounds Monotone DNF Size for k-CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier Analysis on Boolean Functions $\times$ Monotone DNF Size for k-CLIQUE
- **Recorded:** 2026-05-21 06:02 UTC
- **Entry ID:** fd209c52f373

Statement

For any monotone DNF formula on n variables, the sum of absolute values of its Fourier coefficients is $O(\log n)$. For the k -CLIQUE indicator function, this sum is $\Omega(n)$ for $k \geq 3$.

Rationale

Fourier coefficients capture the influence of variables in DNF formulas. Monotone k -CLIQUE requires exponentially many terms, leading to large coefficient sums, while polynomial-size DNFs have logarithmic sums due to sparsity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.04s

```

', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}

```

TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture': 'Fourier coefficient sum is 1'}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture': 'Fourier coefficient sum is 1'}
RESULT: SUPPORTED mean=0.9333333333333333 std=0.24944382578492943 support_fraction=1.0

Judge reasoning

All trials used $n=1$, making the Fourier coefficient sum trivially 1. This violates scaling requirements for both $O(\log n)$ and $\Omega(n)$ claims. | next: Test with $n \geq 1000$ and varying k to observe scaling behavior of Fourier coefficient sums

Real Radical Dimension Bounds AC^0 Circuit Size for PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry $\times$ AC^0 Circuit Size for PARITY
- **Recorded:** 2026-05-21 06:31 UTC
- **Entry ID:** 2c022a5d9d33

Statement

For any AC^0 circuit C computing PARITY on n inputs, the dimension of the real radical of the ideal generated by C 's truth-table polynomial equations satisfies $\dim(\text{rad}(I_C)) \geq \log_2(\text{size}(C)) - O(\sqrt{n})$.

Rationale

Real radicals capture the algebraic structure of boolean functions; their dimension may encode circuit complexity via elimination theory. PARITY's inherent symmetry could create nontrivial constraints on radical varieties, linking algebraic geometry to circuit size.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 241.79s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results; support fraction cannot be evaluated | next: Run test with extended timeout and verify implementation correctness

Eigenvalue Count and SOS Degree for Max-CUT Approximation

- **Verdict:** INCONCLUSIVE
- **Bridge:** Random Matrix Theory $\times$ SOS Degree for Max-CUT
- **Recorded:** 2026-05-21 06:56 UTC
- **Entry ID:** c0336452dd8d

Statement

For a random graph $G(n, 1/2)$, the minimal SOS degree required to approximate max-CUT within a factor of $0.878 + \varepsilon$ is $\Omega(\sqrt{n})$, where the constant depends on the number of eigenvalues of G 's adjacency matrix in the interval $[-\sqrt{(n \log n)}, \sqrt{(n \log n)}]$.

Rationale

The eigenvalue distribution of random graphs influences the structure of the feasible region for SOS relaxations. Concentration of measure in random matrices suggests that eigenvalues cluster around certain values, which could necessitate higher SOS degrees to capture the necessary precision for approximation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 4, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 3, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 4, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 3, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds': True}

- -STDERR- -

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71ec784e.py", line 71, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71ec784e.py", line 43, in run_trial
    eigenvalues = gaussian_elimination(adjacency_matrix)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71ec784e.py", line 26, in gaussian_elim
    factor_i = 1 / augmented_matrix[i][i]
                ~^~~~~~
```

```
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with `ZeroDivisionError`, preventing reliable evaluation of the conjecture's validity across sufficient instances. | next: Fix the `gaussian_elimination` function to handle singular matrices, then re-run tests with more seeds to validate the conjecture

Prime Density in Arithmetic Progressions and Seed Length of Nisan-Wigderson PRGs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Number Theory × Seed Length of Nisan-Wigderson PRGs
- **Recorded:** 2026-05-21 07:37 UTC
- **Entry ID:** ffe6d4da52cf

Rationale

Free entropy quantifies the ‘non-commutative complexity’ of matrix entries, capturing dependencies that correlate with communication bottlenecks. By framing Disjointness matrices as non-commutative distributions, we expose structural constraints that may lower-bound communication requirements via free probability’s asymptotic freeness properties.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

```
lue': -0.5625, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'chi(M) < 4\sqrt{n}'  
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.625, 'instances_tested': 1, 'conjecture_ho  
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.625, 'instances_tested': 1, 'conjecture_ho  
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.4375, 'instances_tested': 1, 'conjecture_h  
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.4375, 'instances_tested': 1, 'conjecture_h  
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5, 'instances_tested': 1, 'conjecture_hold  
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5625, 'instances_tested': 1, 'conjecture_h  
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5625, 'instances_tested': 1, 'conjecture_h  
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.375, 'instances_tested': 1, 'conjecture_ho  
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5, 'instances_tested': 1, 'conjecture_hold
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a045ef7b.py", line 60, in <module>  
    first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])  
                        ~~~~~~  
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a045ef7b.py", line 60, in <genexpr>  
    first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])  
                        ~~~~~~
```

KeyError: 'seed'

Judge reasoning

Test crashed before completing validation; counterexamples show failure but cannot trust results due to crash | next: Fix seed handling in test code and re-run with sufficient sample size

SOS Moment Matrix Spectral Gap and Max-CUT Approximation Ratio

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded:** 2026-05-21 08:09 UTC
- **Entry ID:** 49361c02c035

Statement

For every Max-CUT instance with n variables, if the degree- d SOS moment matrix M_d has minimal eigenvalue $\lambda_{\min}(M_d) \geq \varepsilon$, then the integrality gap of the d -round SOS relaxation is $\leq 1 - \varepsilon$. Conversely, if the integrality gap exceeds 0.878, then $\lambda_{\min}(M_d) < 1 - 0.878$ for all $d \leq \log n$.

Rationale

The spectral gap of the moment matrix encodes the SOS relaxation's ability to capture low-degree certificates. By linking eigenvalue thresholds to approximation ratios, we expose structural constraints between algebraic geometry and optimization hardness, avoiding algebrization barriers through explicit spectral analysis.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.87s

```
value': 1.0142857142857142, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap exceeds 1 - epsilon=5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0714285714285714, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap exceeds 1 - epsilon=5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0059751773049646, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap exceeds 525/188'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0130870210295446, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap exceeds 23245/5392'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0416666666666667, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap exceeds 5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.004379479878373, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap exceeds 131545706/32251283'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.01875, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap exceeds 13/4'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0042481778622288, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap exceeds 16057508/4367783'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0113157894736842, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap exceeds 587/200'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0110294117647058, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap exceeds 5/2'}
RESULT: FALSIFIED counterexample="Gap exceeds 1 - epsilon" first_fail=1
```

Judge reasoning

parse_fail | next: NONE

Additive Energy of Fourier Coefficients and ACC⁰ Circuit Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Additive Combinatorics × ACC⁰ Circuit Size
- **Recorded:** 2026-05-21 08:19 UTC
- **Entry ID:** 2088c7f83a48

Statement

For every Boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$ in P , the additive energy $E(f)$ of its Fourier coefficients satisfies $E(f) \geq 2^{\Omega(n)}$, whereas for ACC⁰ circuits computing f , $E(f) \leq 2^{O(\log^2 n)}$.

Rationale

Additive energy captures the structure of Fourier coefficients, which reflect the correlation of f with low-degree polynomials. ACC^0 circuits, constrained by bounded-depth, exhibit limited structure, leading to lower additive energy. This could expose a gap between P and ACC^0 via Fourier-analytic methods.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_clfe85aa.py", line 69, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_clfe85aa.py", line 47, in run_trial
    transformed_f = fast_walsh_hadamard_transform(f)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_clfe85aa.py", line 26, in fast_walsh_ha
    v = f[j + m]
        ~^^^^^^
```

IndexError: list index out of range

Judge reasoning

Test crashed during execution, preventing evaluation of the conjecture's validity. | next: Debug the fast_walsh_hadamard_transform implementation to fix index out of range error

Betti Number Sum Lower Bound for SOS Max-CUT Degree

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Topology $\times$ SOS Degree for Max-CUT
- **Recorded:** 2026-05-21 08:33 UTC
- **Entry ID:** b782e0cb2c69

Statement

For a graph G on n vertices, the SOS degree required to approximate Max-CUT with 0.878-approximation ratio is at least the sum of the Betti numbers $\beta_0 + \beta_1$ of G 's clique complex. This bound is tight for complete graphs.

Rationale

Betti numbers encode topological complexity of the clique complex, which may correlate with the algebraic structure of the Max-CUT instance. The SOS hierarchy's degree reflects the need to resolve higher-dimensional interactions, potentially mirroring the topological complexity captured by Betti numbers.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

e': ''}

```
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
RESULT: SUPPORTED mean=1.0666666666666667 std=0.24944382578492943 support_fraction=1.0
```

Judge reasoning

All trials used $n=1$ graphs where Betti numbers are trivial, making the conjecture vacuous. SOS degree is capped at 1, preventing meaningful validation of the lower bound. | next: Test conjecture on graphs with $n \geq 4$ to observe non-trivial Betti number interactions

Asymptotic Dimension and Tseitin Resolution Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Geometric Group Theory $\times$ Resolution Length of Tseitin Formulas
- **Recorded:** 2026-05-21 08:47 UTC
- **Entry ID:** 2f7cf0867b85

Statement

For a graph G with asymptotic dimension d , the resolution length of the Tseitin formula on G is at least $2^{\Omega(d)}$.

Rationale

Asymptotic dimension captures the large-scale geometric complexity of the graph, which may directly influence the length of resolution proofs required to refute the Tseitin formula.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98637aa2.py", line 117, in <module>

```

    result = run_trial(seed)
               ^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98637aa2.py", line 96, in run_trial
    literals, clauses = tseitin_formula(graph)
               ^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98637aa2.py", line 41, in tseitin_formula
    clauses.append([-literals[2*i], literals[2*j]])
               ^^^^^^^^^^^^^^^^^
IndexError: list index out of range

```

Judge reasoning

Test crashed with IndexError before producing valid results | next: Fix the Tseitin formula implementation to handle graph vertex indexing properly

Specht Block Spread of Perm Exceeds Det Under Random Linear Lift

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic combinatorics: Specht-module projections via Murnaghan-Nakayama characters on the regular S_n -representation $\times GL(n^2)$ -orbit spread of perm_n vs det_n in matching-coefficient basis (Mulmuley-Sohoni GCT, monotone setting)
- **Recorded:** 2026-05-21 09:38 UTC
- **Entry ID:** 48b7235cd4ff

Statement

Let $X=(x_{ij})$ be an $n \times n$ matrix of formal variables. For a multilinear degree- n polynomial g in x_{ij} , define the matching coefficient vector $c(g) \in Q^{\{S_n\}}$ by $c_\sigma(g) = \sum_i x_{i,\sigma(i)} g$. For each $\lambda \vdash n$, let P_λ be the projection onto the λ -isotypic component of $Q[S_n]$ using the central idempotent $e_\lambda = (f^\lambda/n!) \sum_\sigma \chi_\lambda(\sigma) \sigma$. Define the Specht block count $\rho(g) = \#\{\lambda \vdash n: \|P_\lambda(c(g))\|^2 / \|c(g)\|^2 > 10^{-9}\}$. Conjecture: $\rho(\text{perm}_n) = \rho(\text{det}_n) = 1$ for all $n \geq 2$; and for every $n \in \{4, 5, 6\}$, for a uniformly random invertible $L \in \{-1, 0, +1\}^{n^2 \times n^2}$, with probability $\geq 1 - 1/n$ we have $\rho(\text{perm}_n(L(X))) \geq \rho(\text{det}_n(L(X))) + 1$.

Rationale

Perm_n 's matching vector is the all-ones (max-entropy) direction in $Q^{\{S_n\}}$; det_n 's is the sign vector — both extremal but on opposite 1-dim irreps. A random linear lift L (the GCT analog of a LIFTING query-to-communication gadget) breaks the $S_n \times S_n$ grading that pins each to a single Specht block. Under random L , the trivial-rep-anchored perm should spread across STRICTLY MORE Specht blocks than the sign-rep-anchored det, because the sign vector's interaction with the regular representation under generic linear lift is more rigid (a Schur-Weyl asymmetry inherited from the Foata bijection inverting maj-inv).

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4e5c65c.py", line 189, in <module>

```

    result = run_trial(seed)
    ~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4e5c65c.py", line 126, in run_trial
    rho_perm_n = sum(projection_on_irrep(identity_matrix(n), lambda_partition=[1]*n) > 1e-
9 for lambda_partition in partitions(n)) / len(partitions(n))
    ~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4e5c65c.py", line 126, in <genexpr>
    rho_perm_n = sum(projection_on_irrep(identity_matrix(n), lambda_partition=[1]*n) > 1e-
9 for lambda_partition in partitions(n)) / len(partitions(n))
    ~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4e5c65c.py", line 79, in projection_on
    f_lambda = Fraction(factorial(n), factorial(lambda_partition))
    ~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4e5c65c.py", line 22, in factorial
    for i in range(2, n + 1):
    ~~~~
TypeError: can only concatenate list (not "int") to list

```

Judge reasoning

The test crashed with a `TypeError` before producing any data (factorial was called on a list instead of an integer), so neither the sanity check nor the main 30-seed trial was evaluated. | next: Fix the bug by computing the hook-length formula for f^λ from the partition (or pass $n=\text{sum}(\lambda)$) and rerun the sanity check for $n \in \{4,5\}$ before launching the random-lift trials.

Specht Character Support Bounds Monotone Permanental Formula Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Specht modules and Murnaghan-Nakayama character theory of the symmetric group $S_n \times \text{Monotone arithmetic formula size for multilinear permanental polynomials (GCT det-vs-perm monotone setting)}$
- **Recorded:** 2026-05-21 09:45 UTC
- **Entry ID:** b5cefcac0023

Statement

For a multilinear permanental polynomial $f = \sum_{\sigma \in S_n} c_f(\sigma) \prod_i x_{i,\sigma(i)}$ in $n \times n$ matrix variables with $c_f \geq 0$, define Specht coefficients $\alpha_\lambda(f) := (1/n!) \sum_{\sigma} c_f(\sigma) \chi_\lambda(\sigma)$ for each partition $\lambda \vdash n$, where χ_λ is the irreducible character (computed via Murnaghan-Nakayama), and the effective Specht support $|\text{supp_eff}(f)| := (\sum_{\lambda} (\dim V_\lambda \cdot \alpha_\lambda)^2) / \sum_{\lambda} (\dim V_\lambda \cdot \alpha_\lambda)^4$ (inverse participation ratio). Conjecture: every monotone permanental arithmetic formula F of size s (binary tree of $\oplus$ gates and $\otimes$ gates that multiply polynomials in disjoint row- AND column-subsets, with nonnegative leaf weights) computing f satisfies $|\text{supp_eff}(f)| \leq s$; consequently any ‘pure’ polynomial like perm_n ($|\text{supp_eff}|=1$, character (n)) or its sign-twin det_n (character (1^n)) requires monotone permanental size ≥ 1 trivially, but every monotone formula F that produces a polynomial f with $|\text{supp_eff}(f)|=k$ must have size $\geq k$. Falsifier: a single monotone permanental formula F with size $s < |\text{supp_eff}(f_F)|$.

Rationale

Mulmuley-Sohoni view perm vs det through representation theory of GL-orbit closures; we drop to the more elementary diagonal S_n action where Murnaghan-Nakayama gives $O(n! \cdot p(n))$ -time exact computation of α_λ , bypassing both the natural-proofs barrier (the invariant is relational/categorical — a participation ratio across irreducible isotypics, not a single truth-table number) and algebrization (the construction is not a polynomial-ring identity in the boolean variables but a representation-theoretic projection on S_n cosets). The bound $|\text{supp_eff}(f)| \leq s$ reflects the intuition that each mono-

tone gate can only construct one new ‘pure character direction’, so a formula must have at least as many gates as the character spread of its output — a quantitative refinement of the Jerrum-Snir style monotone obstruction.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:3232ed274e19727bb02719b0038eeab0dad19] Stability in Topology, Arithmetic, and Representation Theory 2022 - [s2:a386ca523f19c66f4e550fdeff35ca56408af] STACS 2006, 23rd Annual Symposium on Theoretical Aspects of Computer Science, Marseille, France, February 23-25, 2006, P

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ae01e1a.py", line 148, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ae01e1a.py", line 119, in run_trial
    f, chi_table = generate_random_formula(n, s)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ae01e1a.py", line 107, in generate_random_formula
    chi_table = {lam: irreducible_character(lam, beta) for lam in itertools.combinations(range(n), n // 2)}
                  ^^^^^
```

NameError: name 'beta' is not defined

Judge reasoning

The test crashed with a NameError (‘beta’ is not defined) before producing any data, so the pre-registered support condition cannot be evaluated. | next: Fix the bug in generate_random_formula (define/pass ‘beta’ to irreducible_character) and rerun the full 720-trial sweep.